

NIROMAN JOE ARO

Cellular Systems Test Engineer

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SUMMARY

- Wireless Engineer with 6+ years of hands-on industry experience in validating modem with expertise in 3GPP Protocols.
- Skilled in analyzing, debugging, testing, documenting, and developing automation framework for KPI analysis for 5G NR/WiFi technology.
- Experience collaborating with cross-functional teams, field engineers and infra vendors to deliver time sensitive corporate priorities.
- **Technical Initiatives:** RF OTA Testing (TRP/TIS) based on CTIA/PTCRB Req, Signal Integrity, RF/Antenna Design

TECHNICAL SKILLS

Wireless Technologies: NR FR1, NR FR2, LTE, IMS, WiFi, OFDMA, MIMO

Protocols: PHY, MAC, RLC, PDCP, RRC, NAS, SIP, UDP/TCP-IP, LPPe, IEEE 802.11 ac/ax

Tools: QXDM, Apex, NOMAD, Spectrum Analyzer, VNA, Oscilloscope, SDR, Wireshark, iPerf, GitHub, VS Code

Programming & AI: Python, C, LLM prompt engineering, RAG, Amazon QuickSight dashboards

Certification: Certified Wireless Network Administrator (CWNA), **Qualcomm** -5G Associate Certificate, **Besser** - RF Wireless Circuits & Test

WORK EXPERIENCE

Senior Cellular Systems Test Engineer, Qualcomm

Feb 2021-Present

- **Cellular Validation:** Developed and executed comprehensive certification, functional and performance test cases for **Cellular** and **FWA** products, validating OTA and backhaul connection performance to ensure end-to-end modem performance, reliability and stability
- Led modem bring-up, calibration, and supervised field testing for major telecom carriers across multiple markets, collaborating closely with Core Network and RAN vendors to resolve issues through weekly status reviews and joint debugging sessions
- Supported commercialization of first ever **ORAN** deployment and was actively involved in optimizing DISH roaming plan
- **System Optimization:** Conducted **5G NR FR1** and **VoNR/VoLTE** performance analysis to enhance modem efficiency and proposed system design optimizations across **PHY, MAC, RRC**, and **IMS** protocol stacks specifically to improve sleep, SR scheduling & reduce RTP loss.
- **Lab Instrumentation:** Executed, conducted and over-the-air (OTA) RF testing in 3m anechoic chambers to characterize wireless devices and analyzed performance metrics of TX/RX. Used IQ samples to debug at further on PHY layer metrics
- **Automation:** Developed an end-to-end **Python Framework** to automate MECA Robot to execute tests that can simulate real time situation and collect KPIs needed for in-depth analysis reducing the manual effort by almost 80%
- Developed specialized LLM prompts for **AI agents** to analyze DL TPUT logs, guiding the extraction of key KPIs into data frames and identifying root-cause analysis for performance issues
- Currently using AI agents to host a custom RAG that is tailored to answer team requirements streamlining internal lookup process
- **Data Analytics:** Built multiple Amazon **Quick Sight Dashboards** to enable daily log analysis/debugging, boosting team efficiency by **50%**

Software Engineering Development, Federated Wireless

Jun 2020-Feb 2021

- **Predictive Spectrum Analytics:** Developed a **Django** based prototype to assess nationwide CBRS spectrum availability and estimate deployable CBSDs across U.S. Profiled and optimized the code to improve processing time by 35% for client-facing demonstrations
- Built a wrapper on **ITM Propagation Model** in C to compute allowable interference by new CBSD's and allocate EIRP to CBSDs

Software Engineering Intern, Federated Wireless

Jun 2019-May 2020

- Developed a Python-based algorithm using SOAP APIs to communicate with Planet's web server and simulated 6 GHz signal behavior in urban/rural regions to deploy CBSDs
- Automated the computation contour **GeoJSON** and integrated upload/download workflows to **MongoDB** with automatic listing of upload/download failures. Performed regression tests and deployed the solution on an AWS EC2 Instance using SFTP access

ACADEMIC PROJECTS

Indoor RF design and simulation

Nov 2018

- Utilized Ran Plan Professional to model an **LTE** network in an indoor environment incorporating DAS concepts. Maximized coverage using antenna tilting techniques and attenuated signal strength through planning to reduce interference

Programming Projects

Nov 2019

- Deployed two BSTN's and computed Handover Latency, based on factors such as RSRP and channel capacity
- Implemented **OFDM** based algorithm to transmit and receive data with SCS of 15KHz to compute EVM and maximum TPUT.

Open-Source Projects | 5G Protocol Stack and Development Group

Jan 2019-May 2020

- Deployed OAI's Private LTE Network on Ubuntu, using SDR as RF front and operated in LTE Band 5 using Wi-Fi as Core Network
- Utilized Signal Generators to transmit and receive signal with/without clock synchronization for diverse modulation techniques

EDUCATION

Master of Science – Telecommunications Engineering

Aug 2018-May 2020

University of Maryland, College Park.

Bachelor of Technology – Electronics and Communication Engineering

Aug 2012-Apr 2016

Anna University, Chennai, India.